Alejandro D. Mosquera García

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Education

University of Pennsylvania, School of Engineering and Applied Science Expected May 2027
M.S.E., Systems Engineering (Submatriculation). GPA: 3.90/4.00 *Philadelphia, PA*

University of Pennsylvania, College of Arts and Sciences Expected May 2027
B.A., Mathematical Economics and Sociology. GPA: 3.79/4.00 *Philadelphia, PA*

- **Ph.D. and Graduate Goursework:** Microeconomic Theory (Ph.D.)*; Linear Systems Theory*; Graph Neural Networks*; Advanced Linear Algebra; Big Data Analytics; Artificial Intelligence; Pricing Policy.
- **Undergraduate Coursework:** Real Analysis*; Probability; Statistical Inference; Statistical Computing; Public Finance; Macroeconomics of Inequality.

* In progress.

Working Papers

Against the Shadow of Coercion: Exposure, Anticipation, and the Limits of Political Alignment in Coerced States' Trade

Advised by Professor Shresth Garg

- Developed a portfolio-reallocation model in which a threatened target rebalances trade across partners under convex adjustment costs, testing it on a dyadic trade–sanctions panel (TIES and GSDB; 1970–2005; 144 countries) with staggered difference-in-differences (Callaway & Sant'Anna, 2021).
- A falsification framework (matched never-treated placebos, triple differences, event studies, OVB sensitivity) supporting post-threat diversification, anticipation effects at the threat-stage, exposure dependence, credibility amplification, and "black-knight" diversion toward sender-aligned partners, also rejecting redirection toward the target's own allies.

Research in Progress

Coerced Network: Optimal Strategy, State Control, Commodities, and Lobbying in Coerced Trade

Master's Thesis. Advised by Professors Rakesh Vohra and Shresth Garg

- Modeled the sender's instrument choice: the sender harms the target through the world-price change its instrument induces, while domestic lobbies favor measures sparing its own exporters. Empirically proved that senders coerce on the side of trade the target's economy depends on, regardless of the nature of their bilateral relation.
- Built a product-level panel of strategic-commodity trade linked to sanctions, threats, and V-Dem state ownership. Trade structure, not state control, governs how targets reroute, although prices erode more for sanctioned good in high-statism nations in the long-run.
- Developing an inverse-optimization approach to recover the weights targets place on cost, exposure, and political factors, and how they diverge for differeing economies.

Crises and The Body: Alternative Embodied Social Mobility Pathways

Senior Honors Thesis, Sociology. Advised by Professors Petra Todd and Pilar Gonalons-Pons

- Developed a theory of the "embodied alternative mobility": alternative institutions arising from state failure, such as sport academies and beauty pageants, that redirect mobility aspirations toward the body.
- Tested it on Argentina's 2001–02 collapse (EPH surveys, a mined city-level academy exposure, dropout panels, a province-year Latinobarómetro trust index) with a gender-based falsification test and wild-cluster bootstrap inference; find more professional football debuts where bureaucratic public employment contracted, for the cohorts who were of school-age during the crisis.

Research Experience

University of Pennsylvania, Department of Economics, Research Assistant Dec. 2023–Present
Supervisor: Dr. Angel Alvarado, Senior Fellow, Penn Initiative for the Study of Markets *Philadelphia, PA*

- Construct a synthetic control counterfactual to estimate the causal impact of Venezuela's 1999 constitutional reform, cleaning cross-country panel data and pre-treatment matching.
- Construct and analyze a panel dataset of state-level Venezuelan food baskets for a study on statewide inflation heterogeneity in Venezuela.

Behavior Change for Good Initiative, Wharton School, Research Assistant Apr. 2026–Sep. 2026
Supervisors: Professors Angela Duckworth and Katy Milkman *Philadelphia, PA*

- Build reproducible Python and R pipelines for behavioral datasets with 1,000,000+ observations; replicate and extend empirical analyses of mega-studies accepted to QJE and Nature.
- Conduct causal-inference analyses using panel methods, fixed effects, event studies, and robustness checks, producing publication-ready tables and figures.

United Nations IOM & PDRI/DevLab, Research Fellow May 2025–Apr. 2026
Remote/Washington, D.C.

- Conduct quantitative policy evaluation of how legal status, reintegration programs, and local institutional conditions shape the integration trajectories of Colombian returnees.
- Consolidate and analyze a dataset of 100,000+ observations spanning 8 years into a comprehensive quantitative report on returnee conditions and demographics.

Penn Development Research Initiative (PDRI/DevLab), Research Assistant Feb. 2025–Apr. 2026
Supervisors: Professors Heather Huntington and Erik Wibbels *Philadelphia, PA*

- Only undergraduate in the Research-to-Policy Student Initiative: translated doctoral-level applied economics research into a policy brief, turning econometric results and causal mechanisms into original policy recommendations.
- Conduct systematic qualitative analysis of focus groups and key informant interviews using structured coding frameworks and AI-assisted text analysis, identifying patterns in institutional performance, forest governance, and community incentives.
- Develop trilingual literature reviews on human trafficking, integrating cross-country empirical findings and policy evaluations.

Ecoanalítica, Economics Research Intern Jul. 2025–Aug. 2025
Caracas, Venezuela

- Authored several of the firm's bi-monthly research white papers for 400+ international and local entities.
- Performed macroeconomic forecasts on oil and currency markets for Latin America and Venezuela; reported weekly findings to the Latin American Alliance of Economic Consulting Firms (LAECO).

Green Swan Capital & Penn Carey Law, Economics Researcher Oct. 2024–Feb. 2025
Philadelphia, PA

- Constructed a panel time-series database combining oil prices, Venezuelan production, sanctions, and GHG emissions.
- Estimated multivariate regression models quantifying sanction exposure and emissions outcomes, and built an empirical policy-risk assessment of Venezuela's oil sector.

Policy Briefs

Welfare Effects of Tuition Price Regulation Feb. 2026
PDRI-DevLab Research-to-Policy Student Initiative Policy Brief

- Prepared with the study's author, Advait R. Aiyer, translating the working paper "Price Regulation in Centralized College Admission Systems" (tuition regulation, enrollment, and welfare in India's centralized engineering-college admissions) into policy recommendations.

Presentations

Against the Shadow of Coercion

Poster presentation, University of Pennsylvania

- CURF Fall Research Expo, Center for Undergraduate Research and Fellowships — Sep. 2026
- Undergraduate Research Poster Exhibit, College of Arts & Sciences Family Weekend — Oct. 2026

Grants, Awards, and Memberships

Kanta Marwah College Alumni Society Undergraduate Research Grant in Economics (USD 1,500) — 2026

Philomathean Society (oldest continuously existing U.S. literary society); Chair, Professor Tea Series — 2025

POLIS (Pi Lambda Sigma) Fellow — 2025

Writer and Editor, Economic Almanac — 2024

Research, Innovation and Entrepreneurship Program — 2023–2024

Policy and Leadership Experience

Friedrich Ebert Foundation, Intern — May 2025–Jul. 2025

Caracas, Venezuela

- Advised the directive board on political action and economic outlooks in Latin America; performed data analysis and literature reviews for public policy publications.
- Crafted economic and public policy programs for congressional and local candidates.

Observatorio Venezolano de Finanzas, Economic Researcher — May 2024–Aug. 2024

Caracas, Venezuela

- Built independent inflation indices from regional food-basket prices, providing an alternative to the distorted official metrics of the Central Bank of Venezuela.
- Estimated the heterogeneity of inflation across Venezuelan regions, comparing regional price dynamics with the national aggregate.

Venezuelan National Interim Government, Educational Policy Contributor — Sep. 2021–Dec. 2021

Barquisimeto, Venezuela

- Revised and drafted policy proposals for the Education Section of the National Salvation Plan, covering nutrition, transportation, curricula, and health programs.

Venezuelan High School Student Union, National Advisor — Jan. 2019–Present

Barquisimeto/Caracas, Venezuela

- Advise 11+ state chapters on political action; organized marches of up to 3,000 participants; introduced student-fare, infrastructure, and funding proposals.

Skills

Programming: Python, R, Stata, SQL (proficient); Qualtrics (intermediate); LaTeX.
Languages: Spanish (native); English and French (fluent); Portuguese (proficient); German and Russian (beginner).